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BATCH 45

ASS 6.3

Task Description #1: Classes (Student Class)

Scenario

You are developing a simple student information management module.

Task

• Use an AI tool (GitHub Copilot / Cursor AI / Gemini) to complete a Student class.

• The class should include attributes such as name, roll number, and branch.

• Add a method display_details() to print student information.

• Execute the code and verify the output.

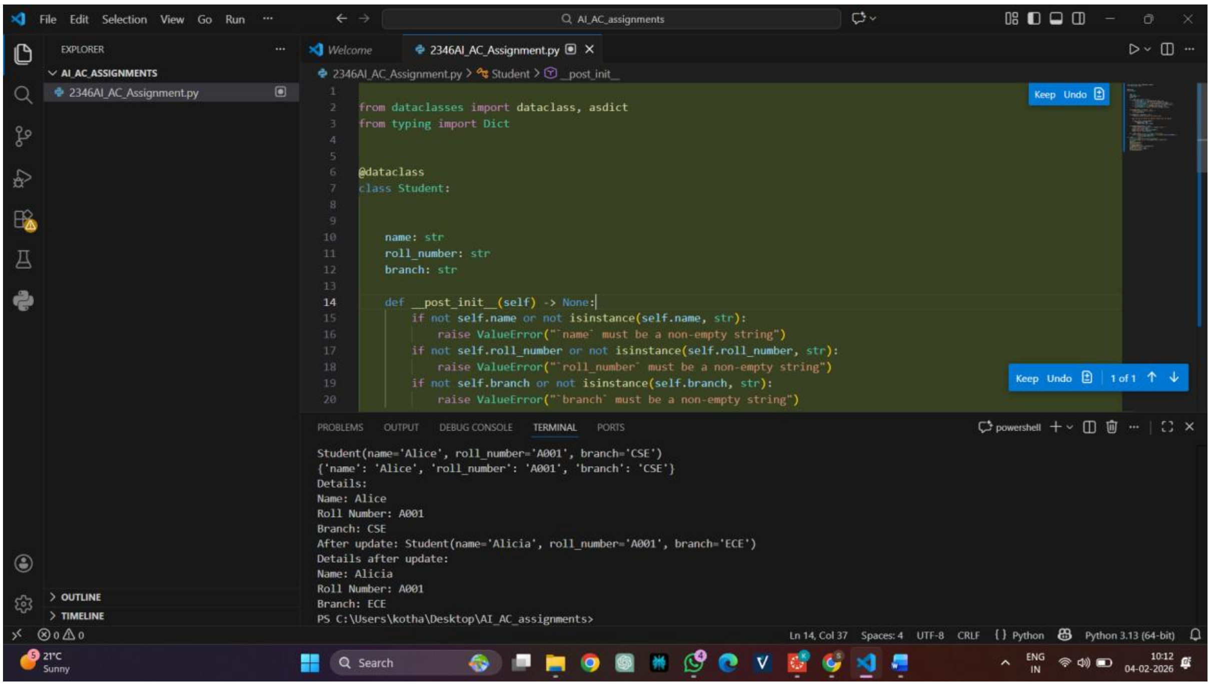
• Analyze the code generated by the AI tool for correctness and clarity.

Expected Output #1

• A Python class with a constructor (__init__) and a display_details() method.

• Sample object creation and output displayed on the console.

• Brief analysis of AI-generated code.



Task Description #2: Loops (Multiples of a Number)

Scenario

You are writing a utility function to display multiples of a given number.

Task

• Prompt the AI tool to generate a function that prints the first 10 multiples of a given number

using a loop.

• Analyze the generated loop logic.

• Ask the AI to generate the same functionality using another controlled looping structure (e.g.,

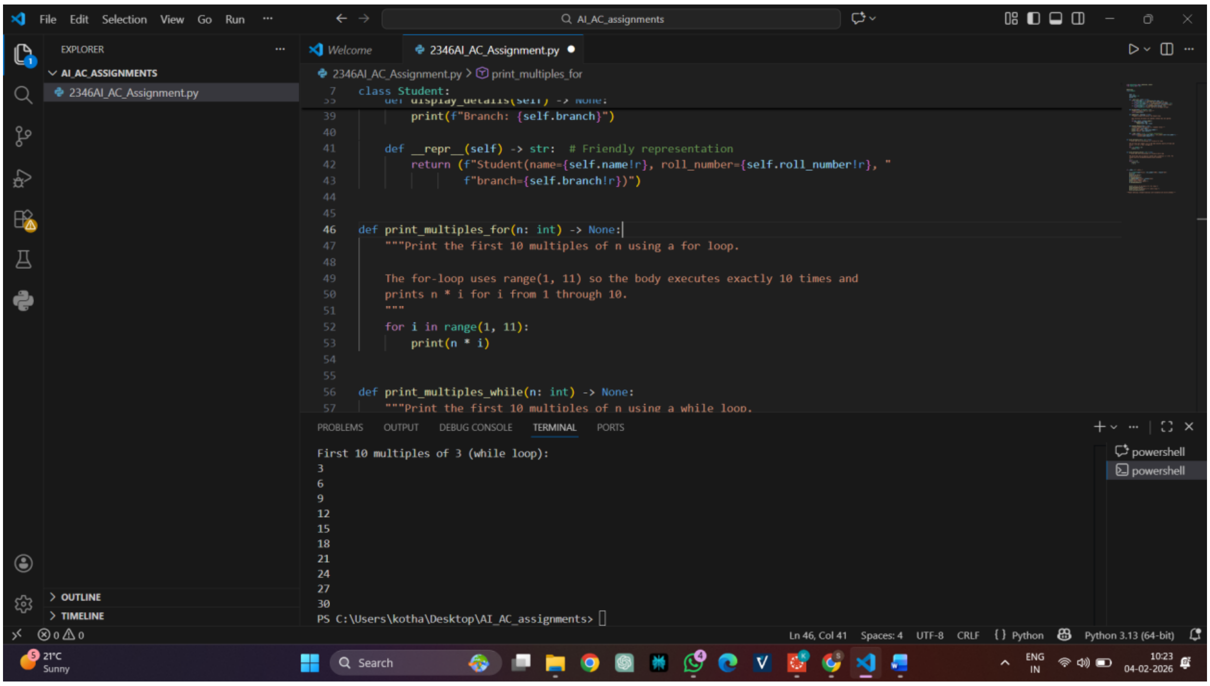
while instead of for).

Expected Output #2

• Correct loop-based Python implementation.

• Output showing the first 10 multiples of a number.

• Comparison and analysis of different looping approaches



Task Description #3: Conditional Statements (Age Classification)

Scenario

You are building a basic classification system based on age.

Task

- Ask the AI tool to generate nested if-elif-else conditional statements to classify age groups

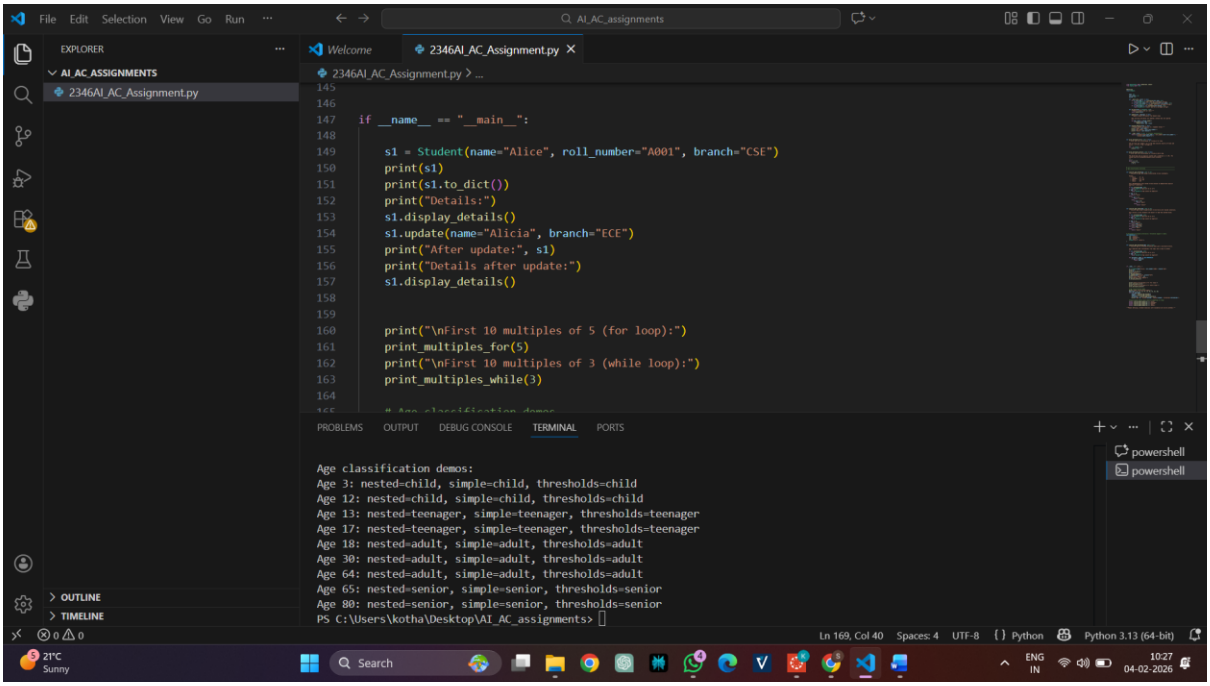
(e.g., child, teenager, adult, senior).

- Analyze the generated conditions and logic.
- Ask the AI to generate the same classification using alternative conditional structures (e.g.,

simplified conditions or dictionary-based logic).

Expected Output #3

- A Python function that classifies age into appropriate groups.
- Clear and correct conditional logic.
- Explanation of how the conditions work.



Task Description #4: For and While Loops (Sum of First n Numbers)

Scenario

You need to calculate the sum of the first n natural numbers.

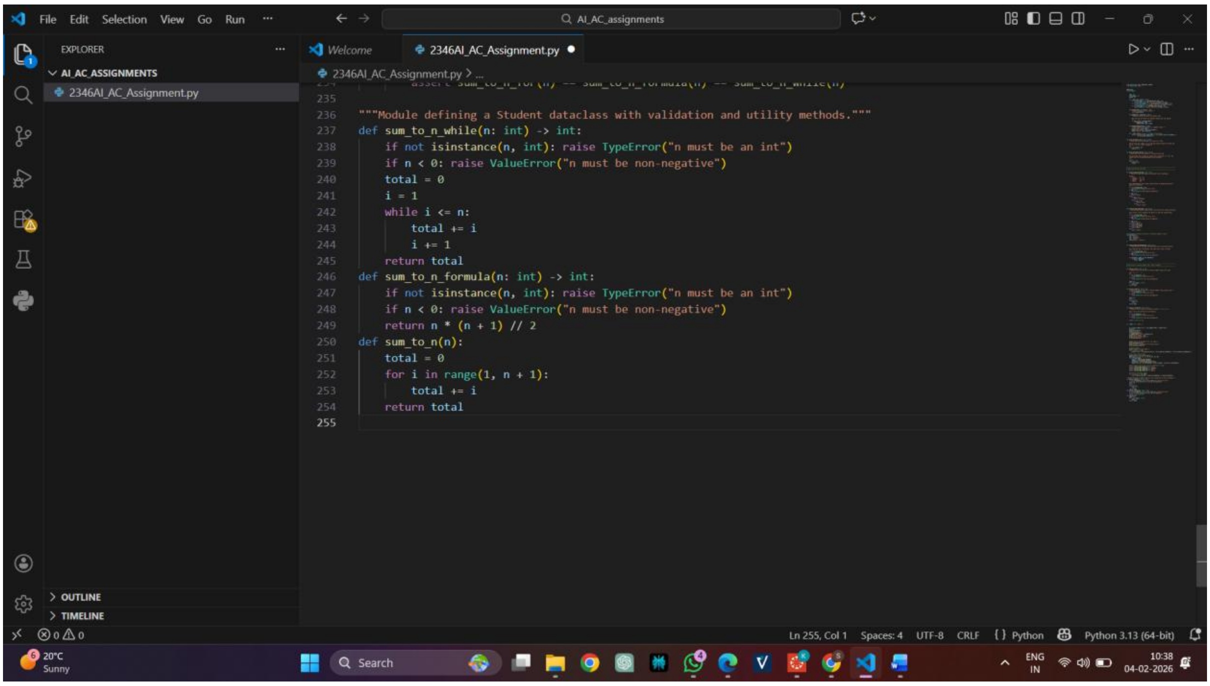
Task

- Use AI assistance to generate a sum_to_n() function using a for loop.
- Analyze the generated code.
- Ask the AI to suggest an alternative implementation using a while loop or a mathematical

formula.

Expected Output #4

- Python function to compute the sum of first n numbers.
- Correct output for sample inputs.
- Explanation and comparison of different approaches



T ask Description #5: Classes (Bank Account Class)

Scenario

You are designing a basic banking application.

Task

• Use AI tools to generate a Bank Account class with methods such as deposit(), withdraw(),

and check_balance().

• Analyze the AI-generated class structure and logic.

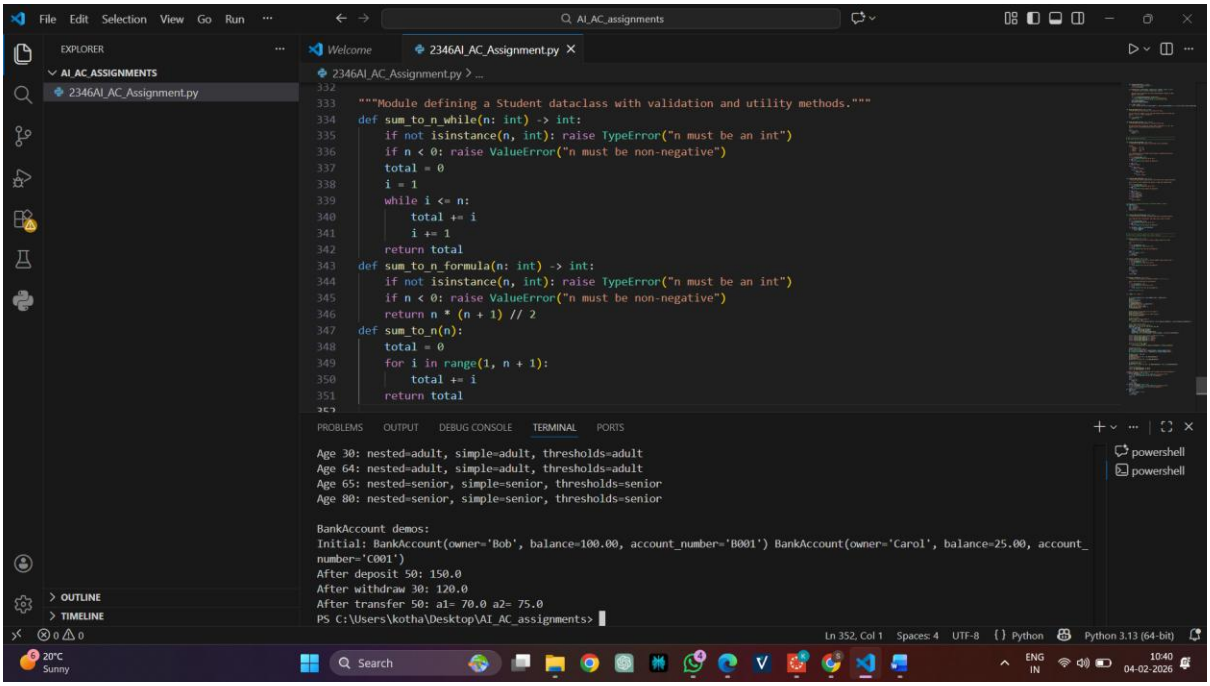
• Add meaningful comments and explain the working of the code.

Expected Output #5

• Complete Python Bank Account class.

• Demonstration of deposit and withdrawal operations with updated balance.

• Well-commented code with a clear explanation


File Edit Selection View Go Run ···
AI_AC_assignments
EXPLORER
AI_AC_ASSIGNMENTS
2346AI_AC_Assignment.py
Welcome
2346AI_AC_Assignment.py
2346AI_AC_Assignment.py > ...
"""Module defining a Student dataclass with validation and utility methods."""
def sum_to_n_while(n: int) -> int:
    if not isinstance(n, int): raise TypeError("n must be an int")
    if n < 0: raise ValueError("n must be non-negative")
    total = 0
    i = 1
    while i <= n:
        total += i
        i += 1
    return total
def sum_to_n_formula(n: int) -> int:
    if not isinstance(n, int): raise TypeError("n must be an int")
    if n < 0: raise ValueError("n must be non-negative")
    return n * (n + 1) // 2
def sum_to_n(n):
    total = 0
    for i in range(1, n + 1):
        total += i
    return total
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Age 30: nested=adult, simple=adult, thresholds=adult
Age 64: nested=adult, simple=adult, thresholds=adult
Age 65: nested=senior, simple=senior, thresholds=senior
Age 80: nested=senior, simple=senior, thresholds=senior
BankAccount demos:
Initial: BankAccount(owner='Bob', balance=100.00, account_number='B001') BankAccount(owner='Carol', balance=25.00, account_number='C001')
After deposit 50: 150.0
After withdraw 30: 120.0
After transfer 50: a1= 70.0 a2= 75.0
PS C:\Users\kotha\Desktop\AI_AC_assignments>
powershell
powershell
OUTLINE
TIMELINE
Ln 352, Col 1 Spaces: 4 UTF-8 Python Python 3.13 (64-bit)
20°C Sunny
Search
ENG IN
10:40
04-02-2026